

MIAE 221: Materials Science for Engineers

Part 2: Chemical Bonding, Potential Energy Wells & Physical Properties

1. The Three Primary (Strong) Chemical Bonds

Primary bonds are strong interatomic attachments involving valence electron transfers or sharing, with bond energies typically ranging from 100 to 1000 kJ/mol (1 to 10 eV/atom).

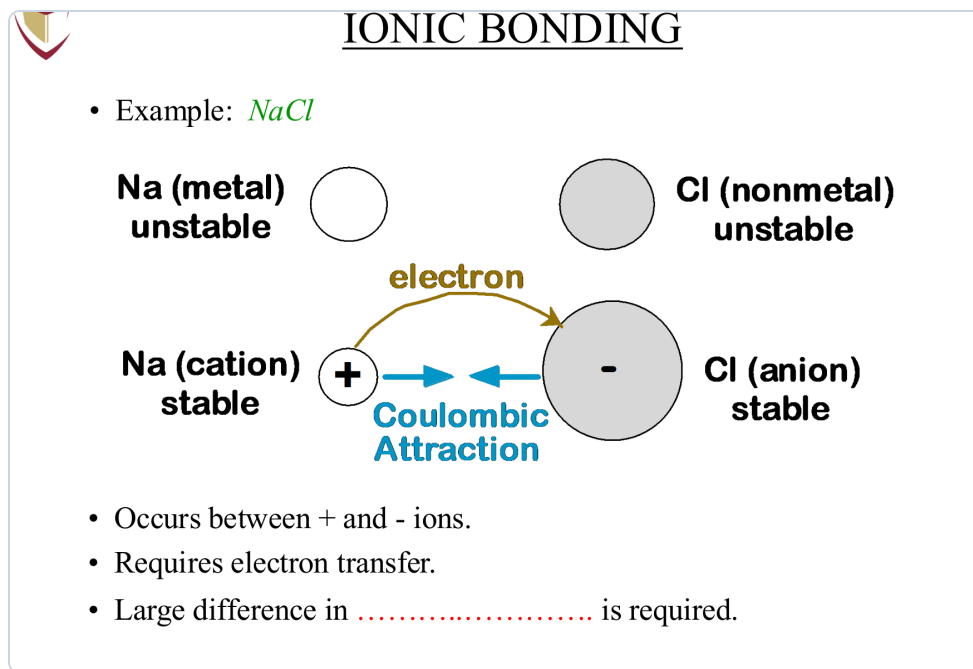


Figure 2.1: Ionic bonding mechanism: Complete electron transfer from electropositive metal cation (Na^+) to electronegative nonmetal anion (Cl^-), forming non-directional Coulombic attraction.

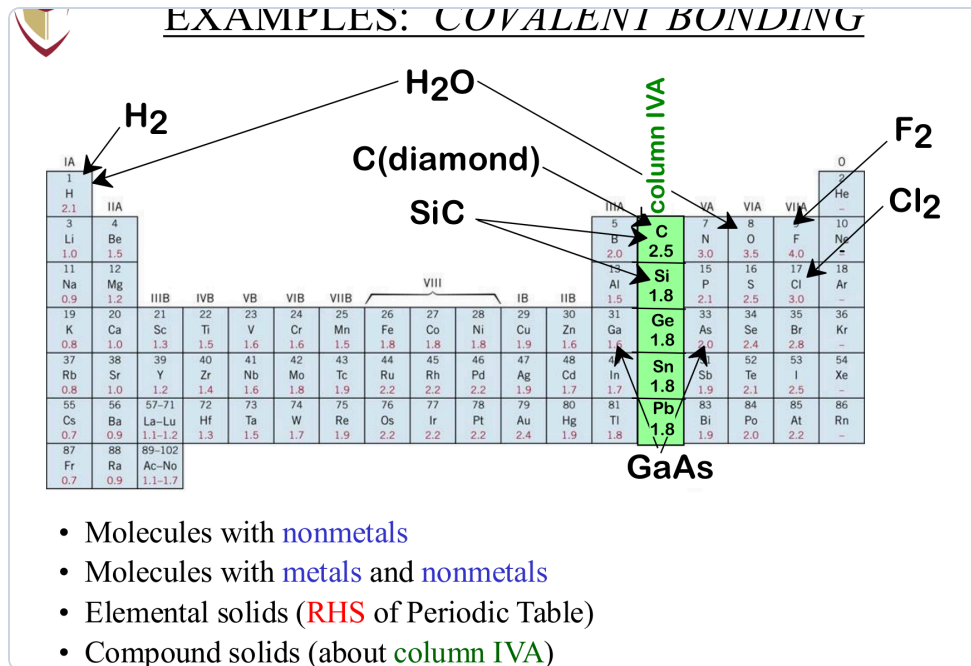


Figure 2.2: Covalent bonding mechanism: Localized orbital overlap and sharing of valence electrons along fixed geometric bond angles.

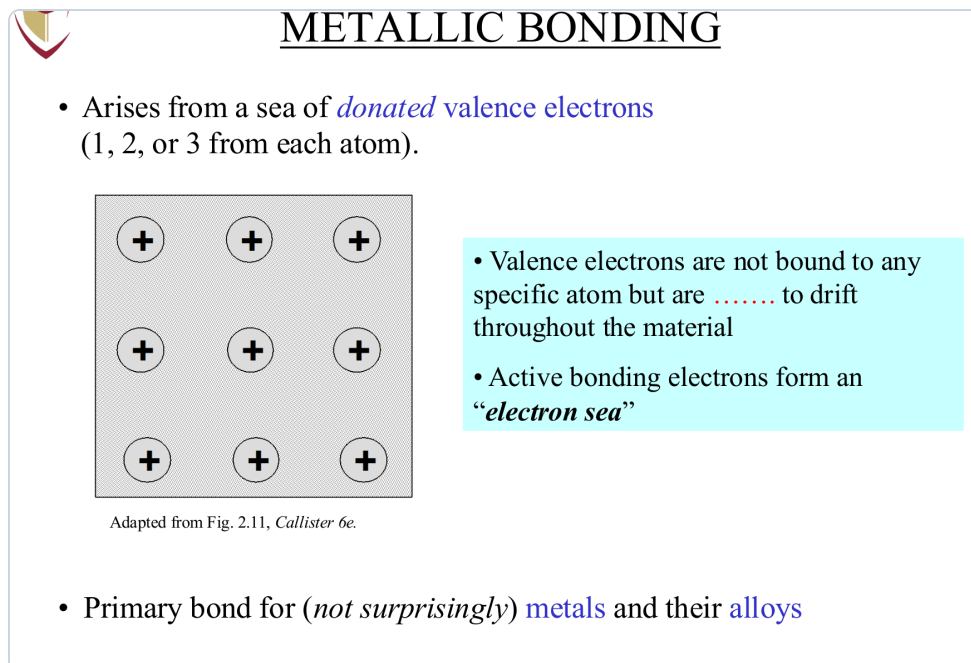


Figure 2.3: Metallic bonding mechanism: Positively charged ion cores bathed in a delocalized, freely mobile electron gas (“sea of valence electrons”).

A. Ionic Bonding (Electrostatic Attraction)

- Physical Mechanism:** Occurs between atoms with large differences in electronegativity ($\Delta X > 1.7\text{--}2.0$), typically a metallic element (electropositive, low ionization energy) and a nonmetallic element (electronegative, high electron affinity).



Example: Sodium ($Na : [Ne]3s^1$) transfers its outer electron to Chlorine ($Cl : [Ne]3s^23p^5$), producing mutually stable octets: $Na^+([Ne])$ and $Cl^-([Ar])$.

- **Directionality: Non-directional.** Because an electric field radiates symmetrically in all spherical directions, a positive ion attracts negative ions equally from every orientation in 3D space. The crystal structure is governed purely by ion size ratios (radius ratio r_c/r_a) and charge neutrality.
- **Coulombic Attractive Force:**

$$F_A = \frac{|z_1||z_2|e^2}{4\pi\epsilon_0 r^2}$$

Where z_1, z_2 are ionic valences, $e = 1.602 \times 10^{-19}$ C, $\epsilon_0 = 8.854 \times 10^{-12}$ F/m (permittivity of vacuum), and r is interionic separation.

B. Covalent Bonding (Electron Sharing)

- **Physical Mechanism:** Occurs between atoms with small differences in electronegativity located near each other on the right side of the periodic table (e.g., $C, Si, Ge, O_2, N_2, CH_4$, and polymer chains). Rather than giving up electrons, adjacent atoms share pairs of valence electrons such that each participating atom achieves a stable, closed-shell inert gas configuration.
- **Directionality: Highly Directional.** Covalent bonds form strictly along specific angles defined by quantum orbital overlap (e.g., sp^3 hybrid orbitals form a strict 109.5° tetrahedral geometry in diamond and silicon).
- **Macroscopic Impact:** Because atoms cannot shift without breaking rigid directional orbital bonds, covalent crystals like diamond, silicon carbide (SiC), and silicon nitride (Si_3N_4) possess **enormous hardness, high melting temperatures, but no plastic slip**.

C. Metallic Bonding (The Electron Cloud)

- **Physical Mechanism:** Occurs in pure metals and metallic alloys (Fe, Cu, Al, Ti , brass, bronze). Metallic elements have few valence electrons loosely bound to the nucleus. In a solid, these valence electrons detach from individual atoms and become completely delocalized, forming a mobile "sea of electrons" (or Fermi gas) that freely permeates the entire crystal. The positively charged ion cores (nucleus + inner core electrons) are held together by mutual electrostatic attraction to this permeating electron cloud.
- **Directionality: Non-directional.** The sea of electrons shields adjacent ion cores equally in all directions.
- **Engineering Properties Derived from Metallic Bonding:**
 1. **High Electrical & Thermal Conductivity:** The free valence electrons move instantly in response to electric fields or thermal gradients.
 2. **High Ductility & Formability:** When a shear stress is applied, crystal planes slide over one another (dislocation slip). Because the bonding is non-directional, the electron glue simply flows with the moving atoms—the crystal deforms plastically without cleaving.
 3. **Opaque & Lustrous:** Free electrons absorb and re-emit all visible photon wavelengths.

2. Mixed Bonding & Pauling's Percent Ionic Character

In reality, pure ionic and pure covalent bonds represent idealized extremes. Most chemical compounds exhibit **mixed bonding** characterized by partial ionic and partial covalent character.

Linus Pauling established an empirical equation relating the **Percent Ionic Character (%IC)** of an interatomic bond to the difference in Pauling electronegativities (X_A and X_B):

$$\% \text{ Ionic Character} = \left[1 - \exp \left(-0.25(X_A - X_B)^2 \right) \right] \times 100\%$$

Material Pair	Element A (X_A)	Element B (X_B)	$\Delta X = X_A - X_B $	Calculated % Ionic Character	Dominant Bond Nature
NaCl	Na (0.9)	Cl (3.0)	2.1	67.4%	Predominantly Ionic
MgO	Mg (1.2)	O (3.5)	2.3	73.4%	Heavily Ionic Ceramic
GaAs	Ga (1.6)	As (2.0)	0.4	3.9%	Covalent Semiconductor
SiC	Si (1.8)	C (2.5)	0.7	11.5%	Strong Covalent Network
CsF	Cs (0.7)	F (4.0)	3.3	93.4%	Almost Purely Ionic

Reference Pauling Electronegativities

Element	H	C	N	O	F	Na	Mg	Al	Si	Cl	K	Ti	Fe	Zn	Br	Te
X	2.1	2.5	3.0	3.5	4.0	0.9	1.2	1.5	1.8	3.0	0.8	1.5	1.8	1.6	2.8	2.1

Step-by-Step Calculation Examples

1. Titanium Dioxide (TiO_2):

- Electronegativities: $X_O = 3.5$, $X_{Ti} = 1.5 \implies \Delta X = 3.5 - 1.5 = 2.0$
- $(\Delta X)^2 = 2.0^2 = 4.0$
- $\%IC = [1 - \exp(-0.25 \times 4.0)] \times 100\% = [1 - \exp(-1.0)] \times 100\%$
- $\%IC = [1 - 0.3679] \times 100\% = \mathbf{63.2\%}$ Ionic (and 36.8% Covalent).

2. Zinc Telluride ($ZnTe$):

- Electronegativities: $X_{Te} = 2.1$, $X_{Zn} = 1.6 \implies \Delta X = 2.1 - 1.6 = 0.5$
- $(\Delta X)^2 = 0.25$
- $\%IC = [1 - \exp(-0.25 \times 0.25)] \times 100\% = [1 - \exp(-0.0625)] \times 100\%$
- $\%IC = [1 - 0.9394] \times 100\% = \mathbf{6.1\%}$ Ionic (and 93.9% Covalent).

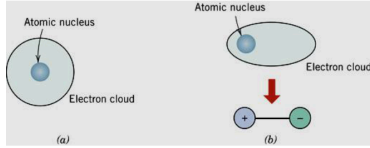
3. Secondary (Weak) Intermolecular Bonds

Secondary bonds (often termed **van der Waals forces**) arise from electrostatic attraction between electric dipoles. They carry bond energies between 4 and 40 kJ/mol (0.04 to 0.4 eV/atom)—roughly an order of magnitude weaker than primary bonds.

Secondary Bonding

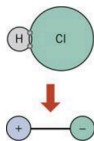
➤ Thermal vibration fluctuations can disrupt charge symmetry which leads to a dipole.

The presence of one dipole can induce a dipole in an adjacent molecule (or atom) and so on.



➤ Polar molecule induced dipoles & bonding

Due to permanent dipole moments in molecules



Unshielded, bare proton in H-O, H-F and H-N bonds lead to strong, hydrogen bonding

- **Hydrogen bonding** is a special case of secondary bonding.
- The **hydrogen bond** is generally stronger than

Figure 2.4: Secondary physical bonding mechanisms (Dr. Medraj MIAE 221 Lecture 3). Left: Fluctuating induced dipoles (van der Waals dispersion forces) in inert gases and nonpolar polymers. Right: Permanent dipole-dipole attractions and Hydrogen bonding in polar compounds (H_2O , HF , NH_3).